

Demo Client · DEMO-0003
11 November 2026 · v1.0

Field	Value
Document	Methodology for monthly investor reporting, attribution and reconciliation
Template version	investor_reporting 1.0.0 / proquant-core 0.1.0
Engagement	DEMO-0003 · Demo Client
Fund	Demo Global Equity Fund, October 2026
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Run ID(s)	ir_20261111T142018Z_287498
Status	Final · 11 November 2026

1. Purpose and scope

The monthly investor pack reports, for each share class of the fund: returns net of fees over standard periods and against the benchmark, risk statistics, the sources of relative performance (sector attribution), portfolio composition, capital activity and assets under management, and the manager's commentary. Every published figure is reconciled with the fund administrator before release. The pack does not re-value the portfolio (the administrator's NAV is the official record), does not give investment advice, and does not forecast returns.

2. Data

2.1 Inputs

Data set	Size	Fingerprint (SHA-256)
nav	2,740 rows	`83ee797ee8778ad7`
capital_activity	256 rows	`af81f041ae034a14`
holdings	2,101 rows	`7c28f355d70becb2`
prices	60,800 rows	`9a49c261f564750a`
benchmark	1,520 rows	`16db9fd05843b6de`
benchmark_sectors	700 rows	`2e08cd97bede3467`
admin_report	126 rows	`d58c7700dd8a46fc`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

2.2 Cleaning rules (all logged in `tables/cleaning_log.csv`)

1. The official NAV is never altered. Exact duplicate NAV rows are dropped; two different NAVs for the same class and day stop the run (to be resolved by the administrator).
2. Share-class names are mapped to the configured classes ignoring case, spaces and punctuation; an unknown class stops the run.

3. Capital flows without a date, amount or a recognised type are dropped; exact duplicates are dropped (to be confirmed with the transfer agent).
4. Exact duplicate holdings lines are dropped; several lines for the same security and date are aggregated. Weights = market value / total of the snapshot (cash included).
5. Price gaps are forward-filled for at most `data.max_ffill_days` business days.
6. Benchmark sector weights are re-scaled to 100% when they miss it by rounding (< 2%); a larger gap stops the run (a sector is missing).

3. Returns

NAV per share is unaffected by subscriptions and redemptions, so returns computed from it are **time-weighted** and **net of all fees** borne by the class:

```
r(t0, t1) = NAV(t1) / NAV(t0) − 1
```

where NAV(t) is the last official NAV on or before t. For a report month M:

Period	Base date t0
1M (month to date)	last day of M-1
QTD	last day of the previous quarter
YTD	31 December of the previous year
1Y / 3Y / 5Y	last day of M-12 / M-36 / M-60
Since launch	the class's launch NAV

Periods longer than one year are annualised geometrically, (1 + r)^(365.25/days) - 1. A period longer than the class history is reported as n/a (never shortened silently). Monthly returns use month-end NAVs; the launch month runs from the launch NAV. The benchmark is measured over the same dates; its since-launch return is computed from each class's launch date. Excess return is the arithmetic difference (percentage points).

4. Risk statistics

Monthly returns over the last `fund.risk_window_months` months (default 36, since launch if shorter; fewer than 12 months are not reported):

Statistic	Definition
Annualised return	geometric: (prod(1 + r))^(12/n) - 1
Volatility	standard deviation of monthly returns x sqrt(12)
Sharpe ratio	(annualised return - rf) / volatility, rf = `fund.risk_free_rate`
Sortino ratio	(annualised return - rf) / downside deviation (monthly returns below rf/12)
Maximum / current drawdown	from daily NAV within the window
Beta, correlation	vs benchmark monthly returns
Tracking error	standard deviation of monthly excess returns x sqrt(12)
Information ratio	(annualised return - benchmark annualised return) / tracking error
Up / down capture	geometric average return in months the benchmark rose / fell, fund / benchmark

5. Attribution (Brinson-Fachler, holdings-based)

For each month t, portfolio sector weights w_p are taken from the previous month-end holdings and sector returns r_p are the buy-and-hold returns of the held securities over the month (cash earns 0). Benchmark weights w_b and returns r_b come from the index provider; R_b = sum(w_b r_b).

```
allocation_s  = (w_p,s − w_b,s) (r_b,s − R_b)
selection_s   = w_b,s (r_p,s − r_b,s)
interaction_s = (w_p,s − w_b,s) (r_p,s − r_b,s)
```

The three effects add up exactly to R_p - R_b, where R_p = sum(w_p r_p) is the holdings-based (gross) return. Months are linked over the year with Carino's coefficients, k_t = (ln(1 + R_p,t) - ln(1 + R_b,t)) / (R_p,t - R_b,t), scaled by the same coefficient for the whole period, so linked effects add up exactly to the compounded excess return.

The difference between R_p and the official NAV return of the class (fees, intra-month trading, cash flows, accruals) is reported as a separate "gross to net" line.

6. Capital activity and AUM bridge

For each class and month: closing NAV = opening NAV + subscriptions - redemptions + investment result. The investment result (P&L net of fees) is the balancing item. The launch month opens with the class's seed capital.

7. Reconciliation and release gate

Check	Comparison	Default tolerance
NAV per share	ours vs administrator	1 bp
Monthly return	ours vs administrator	2 bp (administrators round to 0.01%)
Shares outstanding, total NAV	ours vs administrator	0.01%
Holdings vs NAV	sum of month-end holdings vs total official NAV	0.5%
Benchmark tie-out	sum(w_b r_b) from the sector file vs the index level return	5 bp

A difference outside tolerance that equals a capital flow dealt in the month (or the flows of one day) within `reconciliation.tolerances.explain_abs` is an explained timing difference (WARN). Any other difference outside tolerance is a break (FAIL). With `reconciliation.block_release: true`, a FAIL stamps every document "DRAFT, not for distribution" until the break is resolved and the pipeline is re-run. The reconciliation and sign-off sheet records the analyst, reviewer and fund approvals.

8. Commentary

The market commentary is written by the portfolio manager and reproduced as provided; a missing commentary is shown as a visible placeholder, never generated. The performance commentary is drafted automatically from the computed figures and reviewed by the analyst and the manager before release.

9. Assumptions, limitations and deviations

Official NAV as published (no re-pricing); holdings-based attribution ignores intra-month trading; risk statistics from at most 36 monthly returns; constant risk-free rate; commentary not verified. No deviations from the standard methodology.

10. Reproducibility

Each run folder contains `config_used.yaml`, `audit.json` (analyst, timestamps, code version, input fingerprints, output hashes), `run.log`, every result table as CSV and all documents. Re-running with the same inputs reproduces every figure.

11. References

- CFA Institute (2020). *Global Investment Performance Standards (GIPS) for Firms*.
- Brinson, G., Hood, L., & Beebower, G. (1986). Determinants of Portfolio Performance. *Financial Analysts Journal*, 42(4).
- Brinson, G., & Fachler, N. (1985). Measuring Non-US Equity Portfolio Performance. *Journal of Portfolio Management*, 11(3).
- Carino, D. (1999). Combining Attribution Effects Over Time. *Journal of Performance Measurement*, 3(4).
- Bacon, C. (2008). *Practical Portfolio Performance Measurement and Attribution*, 2nd ed. Wiley.
- Sortino, F., & Price, L. (1994). Performance Measurement in a Downside Risk Framework. *Journal of Investing*, 3(3).